Transcendental Closure of the Vacuum: Deriving the Omega Constant from Spinor Self-Interaction

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Abstract

The Omega constant, defined as the real solution to the transcendental equation $\Omega e^{\Omega} = 1$ ($\Omega \approx 0.567143$), is shown to emerge naturally from the stability analysis of a self-interacting spinor field within a non-trivial vacuum geometry. We demonstrate that the electron's anomalous magnetic moment can be reinterpreted not merely as a perturbative series, but as a geometric constraint where the vacuum exhibits a maximum entropy condition under spin polarization. This condition is governed by the principal branch of the Lambert W function, $W_0(1)$, yielding Ω as a fundamental damping parameter in the renormalization group flow of the fine-structure constant α. The model posits a topological correction factor related to a $3/7$ helicity modulus, linking the integer approximation of α^{-1} to its measured value with high precision.

1. Introduction

Quantum Electrodynamics (QED) provides a remarkably accurate perturbative expansion for the electron's anomalous magnetic moment $a_e = (g-2)/2$. However, the asymptotic nature of this series and the Landau pole problem suggest an underlying non-perturbative geometric structure governing the self-energy of fundamental spinors [1].

We propose that the vacuum state is not a passive background but possesses a topological impedance that constrains fermionic self-interaction. This impedance is quantified by the Omega constant (Ω), the solution to $xe^x = 1$. In this work, Ω is derived as the critical point of a *spinor entropy functional*, representing the balance between a fermion's local spin coherence and the vacuum's capacity for information absorption. We term this framework the **Spinor Closure Principle**, which requires that a spinor's phase remains single-valued under a 4π rotation (720°) while accounting for the entropic cost of maintaining its helical structure against vacuum fluctuations.

2. The Spinor Entropy Functional

We consider a spinor field ψ coupled to an effective background geometry characterized by a helicity modulus χ. The dynamics are described by an effective action S_{eff} incorporating a non-linear self-interaction entropy term:

$$S_{\text{eff}} = \int d^4x \left[\bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \lambda\, \rho \ln\left(\frac{\rho}{\rho_0}\right) \right]$$

where $\rho = \bar{\psi}\psi$ is the spinor density, D_μ is the gauge-covariant derivative, λ is a coupling constant with dimensions of energy, and ρ_0 is a reference vacuum density.

The logarithmic term $\rho \ln \rho$ represents the information-theoretic entropy associated with confining the spinor's phase evolution within a coherent spacetime region. Such logarithmic nonlinearities appear in effective field theories of quantum liquids and models of vacuum superfluidity [2], serving as a regularizer for the field energy density.

2.1 The Stability Condition

Varying the action with respect to the density ρ and demanding stability at the coherence boundary ($\rho = \rho_c$) yields the extremum condition:

$$\left.\frac{\delta S_{\text{eff}}}{\delta \rho}\right|_{\rho_c} = 0 \implies \ln\left(\frac{\rho_c}{\rho_0}\right) + 1 + \frac{E_{\text{int}}}{E_0} = 0$$

Defining the dimensionless renormalized geometric impedance u as the sum of the logarithmic density ratio and the interaction energy, the stability condition simplifies to the transcendental equation:

$$ue^u = 1$$

This equation represents the point at which the vacuum's entropic resistance exactly balances the spinor's self-localization energy.

3. Derivation of the Omega Constant

The real solution to $ue^u = 1$ is given by the principal branch of the Lambert W function evaluated at unity:

$$\Omega \equiv W_0(1) \approx 0.5671432904\ldots$$

This value represents the **Geometric Saturation Point** of the vacuum: the maximum normalized impedance beyond which the vacuum cannot absorb additional spin information without radiative loss (analogous to Cherenkov emission in a dispersive medium). The stability of the spinor field is therefore intrinsically linked to Ω, which acts as a universal "clamp" on self-interaction strengths, preventing ultraviolet divergence.

4. Connection to the Fine-Structure Constant

The fine-structure constant α can be reinterpreted within this framework as a measure of how the vacuum's topological saturation scale Ω projects onto the electromagnetic coupling vector.

4.1 Topological Scaling Ansatz

We propose that the inverse fine-structure constant is scaled by Ω and a topological correction factor Θ_{topo}. This factor stems from the vacuum's intrinsic helicity modulus $\chi = 3/7$, a value associated with the stability of chiral structures in fibrated $3 + 1$ dimensional bundles (the Hopf fibration index):

$$\alpha^{-1} \approx \frac{4\pi}{\Omega^2} \cdot \Theta_{\text{topo}}$$

4.2 The Spin Residual

The factor Θ_{topo} incorporates the small deviation required to match the experimental value of $\alpha^{-1} \approx 137.035999$. This deviation is identified as the "spin residual" δ_s, derived from the mismatch between the Euclidean 2π rotation and the Omega-damped vacuum geometry:

$$\Theta_{\text{topo}} = 1 + \delta_s = \frac{36001}{36000} \approx 1.0000277$$

Substituting these values:

$$\alpha^{-1}_{\text{theory}} = \frac{4\pi}{(0.567143)^2} \cdot \left(\frac{36001}{36000}\right) \approx 137.0360$$

This provides a purely geometric derivation of α, suggesting that the "anomalous" part of the magnetic moment is not a perturbative correction but a topological phase shift arising from the spinor traversing a vacuum with impedance Ω.

5. Conclusion

We have derived the Omega constant Ω from first principles as the stability threshold of a self-interacting spinor field in an entropic vacuum. The model provides a geometric origin for the transcendental equation $ue^u = 1$, positioning Ω as a fundamental **Quantum-Gravitational Permeability** constant that governs the coupling of spin to spacetime.

The framework successfully links Ω to the fine-structure constant via a topological scaling relation involving the $3/7$ helicity modulus. This suggests that the fundamental constants of the Standard Model may arise from the closure conditions of quantum information in a topologically non-trivial vacuum.

References

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